

Pokhara University
Faculty of Science and Technology
(BEE)

1. Course Description

This course covers the various concepts of electrical circuits, theorems as well as the concepts of electrical machines. This course emphasizes on fundamental concept, principles and properties of electrical circuits, circuit parameters and its application. It also covers the concepts of DC and AC electrical circuit analysis and electrical machine.

2. General Objective

The general objectives of this course are:

1. To acquaint the students with AC and DC electric circuits, steady state behavior of single phase and three phase AC electrical circuits.
2. To make the students able to distinguish and use electrical devices and machines.

1. Methods of Instruction

Lecture, Discussion, Readings, Lab works, Project works

1. Contents in Detail

Specific Objectives	Contents
1. Describe the generation, transmission, distribution and use of electrical energy.	Unit I: Introduction (2 hrs) • Role of electricity in modern society, • Energy sources and production, • Generation, transmission and distribution of electrical energy, • Consumption of electricity
1. Evaluate voltages and currents across the electric circuits using voltage and current divider rule. 2. Use Kirchhoff's laws to evaluate current and voltage in DC circuits. 3. Obtain the equations for DC electric circuits using mesh and nodal analyses and solve them to evaluate current(s)/voltage(s). 4. Explain and use network theorems to reduce a DC circuit to a simple equivalent.	Unit II: DC Circuit Analysis (15 hrs) 1. Circuits concepts (lumped and distributed parameters), linear and nonlinear parameter, passive and active circuits, 2. Circuit elements (Resistance, capacitance and inductance), their properties and characteristics in a geometrical and hardware aspects, 3. Color coding, Series of parallel compilation of resistances, Equivalent resistance and its calculation, star-delta transformation, 4. Concept of power, energy and its calculations, 5. Short and open circuit, 6. Ideal and non-ideal sources, source conversion, 7. Voltage divider and current divider formula, 8. Kirchhoff's current and voltage laws,

	9. Nodal method and mesh method of network analysis (without dependent source), 10. Network theorem (Superposition, Thevenin's, Norton's, maximum power transfer theorem)
11. Explain generation of single-phase alternating quantities and its characteristics. 12. Calculate average and root-mean-square values of alternating voltage(s)/current(s). 13. Analyze the steady state behavior of single-phase AC electric circuits.	Unit III: Single Phase AC Circuits Analysis (10 hrs) 1. Generation of EMF by electromagnetic induction, Generation of alternating voltage, 2. Sinusoidal functions-terminology (phase, phase angle, amplitude, frequency, peak to peak value), average values and RMS or effective value of any types of alternating voltage or current waveform, 3. Phase algebra, power triangle, impedance triangle, steady state response of circuits (RL, RC, RLC series and parallel) and concept about admittance, impedance, reactance and its triangle), instantaneous power, average real power, reactive power, power factor and significance of power factor, 4. Resonance in series and parallel RLC circuit, bandwidth, effect of Q factor in resource
5. Explain generation of poly phase alternating quantities and its characteristics. 6. Analyze the steady state behavior of three phase AC electric circuits. 7. Describe the measurement of three phase power.	Unit IV: Poly-phase AC Circuit Analysis (6 hrs) 1. Generation of three phase alternating quantity and Concept of a balanced three phase supply, 2. Differences between single phase and three phase system, 3. Star and delta connected supply and load circuits., Line and phase voltage/current relations, power measurement, 4. Concept of three phase power and its measurement by single and two wattmeter methods
5. Analyze the difference between electric and magnetic circuits. 6. Explain the working principle of single-phase two-winding transformer. 7. Analyze open circuit and short circuit tests of single-phase two-winding transformer. 8. Explain the performance and operation of DC machines 9. Explain the construction, working principle and use of induction motors.	Unit V: Electrical Machines (12 hrs) 1. Differences and similarities between electric circuit and magnetic circuit 2. Transformers: Principle of operations, features, equivalent circuits, efficiency & regulation, open circuit & short circuit tests of single phase two winding transformer. 3. DC generator: Construction features, working principles, basic characteristics 4. DC motors: Performance & operation, basic characteristics, speed control & selection of motors 5. AC machines: Single phase and three phase induction motors (working principles, construction features and uses)

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx

Full marks: 100

Course title: Programming in C (3-1-3)

Pass marks: 45

Nature of the course: Theory & Practical

Time per period: 1 hour

Level: Undergraduate

Total Periods: 45

Program: BE

5. Course Description

This course first introduces the different computer/programming languages- low-level and high-level languages and generation of the programming languages. This course enables the students to solve a given problem using a structured programming language- the C language through the problem-solving steps- problem analysis, design of algorithm, flowcharting, coding the algorithm, executing and compiling/interpreting the program, testing and debugging the program and finally well documenting the program for future understanding. The aim of this course is to provide the basic knowledge of structured programming concepts to develop a program using the C language.

6. General Objectives

The general objectives of this course are:

7. To provide the basic knowledge of computer/programming languages and generations of programming languages.
8. To provide the basic knowledge of problem-solving steps to resolve a particular problem using a computer program.
9. To provide the basic concepts of structured programming with the C language.

10. Methods of Instruction

Lecture, Discussion, Readings, Lab works, Project works

11. Contents in Detail

Specific Objectives	Contents
1. familiar with the various types of programming languages, programming paradigms and software 2. analyze a problem, find the solution, implement the solution using a particular computer/programming language and finally prepare its documentation	Unit I: Programming Languages and Problem Solving (6 hrs) 1.1 Programming Languages- machine-level language, assembly language and high-level language. 1.2 Software and its types 1.3 Generations of programming languages 1.4 Structured programming 1.5 Problem solving using computer- Problems Analysis (understanding of the problem, feasibility

	and requirement analysis), Design (Algorithm and flowchart), Coding (compilation/interpretation and execution), Testing and debugging, Implementation, Evaluation and Maintenance of computer programs, Program documentation
3. explain the history and features of C programming, implement the various constructs of C programming like constants, variables, keywords, data types, operators and formatted I/O functions etc. 4. write and evaluate the expressions in C programs	Unit II: Introduction to C (6 hrs) 2.1 History of C language 2.2 Features of C 2.3 The C as a middle-level language 2.4 The C as a system programming language 2.5 The C character set 2.6 Keywords and Identifiers 2.7 Data types 2.8 Constants, variables and their declaration 2.9 Formatted input/output functions 2.10 The C Operators
5. implement the various control structures of C programming with reference to algorithm or flowchart they developed already	Unit III: Control Structures (6 hrs) 3.1 Introduction and types of control statements- sequential, branching and looping statements 3.2 Branching statements- simple if statement, if-else, nested if, if-else-if ladder and switch statements 3.3 Looping statements- for loop, while loop, do-while loop, nested loop 3.4 The break, continue and goto statements
6. represent/store homogenous type of data as array and realize the multi-dimensional array representation 7. implement string handling functions to handle textual data	Unit IV: Arrays and Strings (6 hrs) 4.1 Introduction to arrays 4.2 One dimensional and Multidimensional arrays 4.3 Initialization of arrays and accessing the elements of arrays 4.4 Strings- the character arrays 4.5 Functions related to the strings
8. use function to perform certain operation 9. use macro definition, header files and function prototype and implement them in their program	Unit V: Functions (6 hrs) 5.1 Introduction 5.2 Importance of functions 5.3 Returning a value from a function and sending a value to a function 5.4 Function prototypes 5.5 Calling a function- Call by value 5.6 Recursive functions 5.7 Passing an array to a function 5.8 Local variables, formal parameters and global variables 5.9 Storage classes 5.10 Pre-processor directives- C libraries, macros and header files

10. implement the pointers in program 11. allocate and deallocate memory dynamically in C language	Unit VI: Pointers (6 hrs) 6.1 Introduction 6.2 Pointer operators 6.3 Pointer arithmetic 6.4 Returning multiple values from functions using pointers 6.5 Pointers and Arrays 6.6 Double indirection 6.7 Dynamic memory allocation
12. able to understand how the heterogeneous data are represented and stored in structure and union and implement them in program	Unit VII: Structure and Union (5 hrs) 7.1 Definition of Structure 7.2 Nested-Structure 7.3 Array of Structure 7.4 Structures and Pointers 7.5 Union 7.6 Self-referential structure
13. create files, access the contents of the file and perform the required operations in the file.	Unit VIII: Files and File Handling (4 hrs) 8.1 FILE pointer, File opening modes (read, write, append) 8.2 File handling functions 8.3 Creating and operating a file in different modes

5. List of Experiments

Laboratory works should cover all the concepts of C language studied in the lectures. Students should submit a final project that uses all the constructs and features of C studied in this course. The marks for the practical work will be based on the project work.

Tools: any C/C++ compiler

6. List of Tutorials

The various tutorial activities that suit this course should cover all the content of this course to give students a space to engage more actively with the course content in the presence of instructor. Students should submit tutorials as assignments or class works to the instructor for evaluation. The following tutorial activities of 15 hrs should be conducted to cover all the content of this course:

14. Discussion-based Tutorials: (2 hrs)
 - a. Evolution of Programming languages and its generations (Class discussion)
 - b. Software and its types.
 - c. Generations of programming languages.
 - d. Structured programming. (Oral Presentation).
15. Problem solving-based Tutorials: (10 hrs)
 - a. Develop algorithms and flowcharts to solve various problems such as to find largest number among three numbers, prime numbers, temperature conversion, product of matrices, finding sum of the terms in series, printing various patterns etc.
 - b. Develop the C programs for the problems for which you developed the algorithms.
 - c. Write a program to pass an array to a function.
 - d. Write a program to use pointers to pass multiple values from a function.

- e. Write a program to use the basic string functions to manipulate string data.
 - f. Write a program to use the principle of recursion to solve the complex problems such as to find factorial of a number, fibonacci series.
 - g. Write a program to illustrate the macros and header files.
 - h. Write a program to illustrate how memory is allocated and deallocated in C language.
 - i. Write a program to use the nested structure. Discuss the scenarios when the structures and unions are used in real practice.
 - j. Write a program to solve simple file handling problems.
16. Review and Question/Answer-based Tutorials: (3 hrs)
- a. Case study of “Development of C with the UNIX operating system and origin of C++ languages” followed by Oral Presentation in class.
 - b. Students ask questions within from the course content and assignments and review key course content in preparation for tests or exams.

7. Evaluation System and Students’ Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	50	Theory		30
		Attendance and Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Practical		20
		Attendance and Class Participation	10%	
		Lab Report/ Project Report	20%	
		Practical Exam/ Project Work	40%	
		Viva	30%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all

the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

8. Prescribed Books and References

Text Books

17. Gottfried, B. *Programming with C*. New Delhi: McGraw-Hill.
18. Balagurusamy, E. *C programming*. New Delhi: McGraw-Hill.

References

1. Kelley A. & Pohl, I. *A Book on C, Programming in C*, USA: Addison-Wesley.
2. Kernighan B. W. & Ritchie, D. M. *The C Programming Language*. USA: Prentice Hall Publication.

Pokhara University
Faculty of Science and Technology

Course Code.: xxx xxx
Course title: Computer Workshop (0-0-3)
Nature of the course: Practical
hour
Year: First
Level: Bachelor

Full marks: 100
Pass marks: 45
Time per period: 1

Total periods: 30
Program: BE

3. Course Description

This course provides the knowledge and hands-on skills of computer hardware, software, computer networking and enables students to identify and rectify the onboard computer hardware, software and network related problems. Students will be able to understand the hardware specifications for the operating system and various application programs.

General Objectives

The general objectives of this course are:

4. To familiarize the students with the computer systems hardware, basic practical works and computer networking concepts
5. To make the students competent in install/update operating system and various application software, manage data backup and restore operations on a computer system

6. Methods of Instruction

Specific instructional techniques: Lab works, Project works

7. Contents in Detail

Specific Objectives	Contents
8. Recognize the computer hardware and accessories.	Unit 1: Introduction to Desktop/Workstation Assembly (2) 1.1 Demonstration of computer hardware and peripheral devices: Processors, Motherboard, Memory, Storage devices, I/O Devices. 1.2 Demonstration of Power Supply-SMPS, Internal cablings and Ports.
9. Assemble and Disassemble PC with power supply, internal cabling, Motherboard and PCI Device	Unit 2: Assembling Desktop Computer (2) 2.1 Assembling steps and precautions. 2.2 Setting of motherboards, memory, hard disk, processors, CDROM, SMPS and other devices.
10. Assemble Laptop power supply, internal cabling, Motherboard and PCI Devices.	Unit 3: Assembling Laptop (2) 3.1 Assembling steps and precautions. 3.2 Demonstration of motherboards, memory, hard disk, processors, CDROM, SMPS and other devices.
11. Install OS (Linux and Windows) in Desktop/Laptop	Unit 4: Laptop/Desktop OS Installation (6) 4.1 BIOS setting, firmware types, BOOT configuration.

	4.2 Hard disc partitioning. 4.3 OS installation (Linux, windows). 4.4 User account management a. Printer, Scanner installation.
12. Recognize computer networks-wired LAN and Wireless LAN configuration.	Unit 5: Networking and Internet Setup (5) 5.1 Introduction to Computer Network, Network topologies, wired and wireless networking media. 5.2 Network cabling, cable types and connectors. 5.3 Basic wireless AP configuration.
13. Install and update application software and utility software. 14. Scan and remove viruses from computer system.	Unit 6: Software Installation (5) 6.1 Installation of application and utility software. 6.2 Update of firmware patches, 6.3 Virus scanning.
15. Setup Server of Mail/Data/Domain/FTP. 16. Apply FTP/SCP Client tool to upload/download files to FTP server.	Unit 7: Server Installation and Backup (3) 7.1 Introduction to server. 7.2 IP Address setting. 7.3 Study of different server (Email, Data, Domain, FTP).
17. Use the basic troubleshooting tools and utilities.	Unit 8: Hardware and Software Troubleshooting (5) 8.1 Basic Repair and Maintenance of Desktop/Laptop. 8.2 Replacement of Passive Components. 8.3 Uses of basic software troubleshooting commands. 8.4 Uses of trouble shooting tools and utilities.

18. Evaluation system and Students' Responsibilities

Evaluation System

The evaluation of a student may consist of assignments, attendance, test-exams, term-exams, lab reports and projects etc. The tabular presentation of the evaluation is as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End examination	0	Attendance & Class Participation	5%	100
		Assignments	10%	
		Quizzes/MCQ	10%	
		Project Work	10%	
		Viva	15%	
		Term Exam	50%	
	0	Total Internal		100

Full Marks: 0 + 100 = 100

Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

19. Prescribed Books and References**References**

20. Rosch, W. R. *The Winn L. Rosch Hardware Bible*. USA: Sams.
21. Bose, A. K. *Hardware and Software of Personal Computers*. New Delhi: New Age International.
22. Rajaraman, V. and Adabala, N. *Fundamental of Computers*. New Delhi: PHI Learning.

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Course No.:	Full Marks: 100
Course Title: Communication Techniques (2-2-0)	Pass Marks: 45
Nature of the Course: Theory	Time per Period: 1
hour	
Year: First	Total Periods: 30
Level: Undergraduate	
Credits: 2	Program: BE

1. Course Description

This course is designed to offer a comprehensive introduction to first-year engineering students to strategies that will help them create effective technical documents and presentations.

It covers the entire gamut of technical communication in thirteen major parts, namely, thinking about the audience, purpose, and genre; ethical issues; researching; technical writing; designing documents; visual communication; communication in the workplace; writing proposals; reporting information; writing how-to documents; reporting document usability; taking communication online; and presenting communication orally.

This course takes a rhetorical approach to technical communication. This means that instead of setting up a list of rules that students should apply uniformly to all writing situations, this course introduces them to the bigger picture of how the words they write can affect the people intended to use them. By understanding who the readers or users are and what they need in a technical document, students can adapt their knowledge to their situations to provide them with what they need and this will facilitate how to educate themselves as well as facilitate the learning of others. There are also several different kinds of exercises and assignments. in-class exercises are short assignments intended to be done by students in class in 15 or 20 minutes. They ask students to use the main ideas discussed in the previous section and to think critically about those ideas. There are also assignments and major projects, and these group activities further learning through opportunities to work with others, to discuss the course content, and to hear others' bright ideas that might spark greater creativity overall.

2. General Objectives

The objectives of the course are to enable students to

2. practice technical writing strategies
3. apply the concepts that they need to make good decisions about how to write a document
4. guide them in developing a good working draft
5. test their document with members of the target user group and receive constructive feedback
6. present usable information on a specialized (and usually technical) subject
7. write technical proposals, reports, and documents
8. make oral technical presentations
9. communicate in the workplace.

3. Methods of Instruction

In this course the idea is that students should read the chapter prior to class and, when they get to class, the teacher can quickly summarize for them the main points to which they should pay attention. Then the teacher can assign an exercise that will reinforce the theoretical concept and get students started working with it. There are also several different kinds of exercises and assignments: in-class exercises are short assignments intended to be done by students in class in 15 or 20 minutes, and here students use the main ideas discussed in the previous section and think critically about those ideas; lab assignments are slightly longer than in-class exercises, but they are still intended to be completed (or at least worked on) during a class meeting; the third type of assignment is the major project, which is a longer, more formal assignment that has students consider the main concepts from the chapter (or chapters) to produce an effective example of one of the main genres of technical communication. The teacher can assign the major project at the beginning of a particular unit of study and have it due the following week or at the end of the term, depending on the teacher's course plan and schedule. It is also important to have the students work collaboratively on the in-class and lab assignments. Depending upon the teacher's course goals, they may decide to cover the chapters in the order in which they are presented in the syllabus but they can use the chapters in the order that suits their students' needs and the assignment schedule.

4. Contents in Detail

The course is divided into eight units of two weeks each.

Weeks	Specific Objectives	Course Contents
1-2	Introducing the basic concepts of audience, purpose, and genre; Introducing ethical issues and conducting research	Thinking about audience, purpose, and genre; Leading and misleading the reader: 10. ethics at work 11. ethics for students 12. how is ethics related to technical communication? 13. researching technical subjects
3-4	Writing for the workplace; Style in technical prose	Writing email and letters for the workplace; 14. writing messages: email, memos, letters 15. writing messages: informative, positive, negative, persuasive 16. how is ethics related to technical communication Writing technical prose: 17. clarity, cohesion, conciseness, parallelism
5-6	Designing technical documents; Incorporating visuals into technical documents	Designing documents and page layout; Communicating through visuals
7-8	Writing effective proposals; Writing instructional documents	Writing winning proposals; Writing how-to documents: 18. instructions, procedures, and manuals
9-10	Testing instructional documents; Reporting technical information, part 1	Testing and reporting document usability; Reporting technical information: 19. recommendation reports
11-12	Reporting technical information,	Reporting technical information:

	part 2; reporting technical information, part 3 and accessing technical documents online	20. white papers or information reports; Taking technical communication online: 21. sharing documents electronically
13-14	Writing documents to be used online; Status or progress reports	Taking technical communication online: 22. writing online documents; Reporting technical information: 23. status or progress reports
15-16	Presenting technical information orally; Students present some aspect of their major projects	Oral presentation of technical reports; Student presentations of major project

5. Evaluation System and Students' Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, project work, class participation, etc. The tabular presentation of the internal evaluation is as follows.

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End Examination	50	Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

6. Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Book

Graves, H., & Roger, G. *A Strategic Guide to Technical Communication*. 2nd ed., London: Eurospan Group, 2012.

References

Greenlaw, R. *Technical Writing, Presentation Skills, and Online Communication: Professional Tools and Insights*. IGI Global, 2012.

Gurak, L. J. & John, M. L. *Strategies for Technical Communication in the Workplace*. 2nd ed., Pearson, 2013.

Kmiec, D. & Bernadette, L. *The IEEE Guide to Writing in the Engineering and Technical Fields*. Wiley, 2017.

Markel, M. *Technical Communication*. 11th ed., Bedford/St. Martins, 2015.

Mirel, B. & Rachel, S. eds. *Reshaping Technical Communication: New Directions and Challenges for the 21st Century*. Lawrence Erlbaum, 2002.

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx
Course title: Calculus I (3-2-0)
Nature of the course: Theory
Level: Undergraduate

Full marks: 100
Pass marks: 45
Time per period: 1 hour
Total Periods: 45
Program: BE

1. Course Description

This course is designed for developing competency of the students on the fundamental concepts, principals and applications of differential and integral calculus for solving engineering problems. It is equipped with differential calculus, integral calculus and ordinary differential equations. The review part of the content is based on previous learning in the school level. The course will be delivered through lecture method, assignment on practically base engineering problems and class tests.

2. General Objectives

The course is designed with the following general objectives:

- To acquaint the students with applications of differential and integral calculus in engineering.
- To expose the students with the application of differential equations for modeling engineering problems.

3. Methods of Instruction

Lecture, tutorials and assignments

4. Contents in Detail

Specific objectives	Contents
• Explain the importance of limit and continuity in differential problems and use Leibnitz theorem to evaluate higher derivatives.	Unit I: Limit Continuity and Derivatives (5 hrs) 1.1 Introduction 1.2 Limit, continuity and differentiability 1.3 Higher order derivatives by Leibnitz method.
• Apply derivatives in mean value theorem, series expansion, asymptotes and trace curve for the given function.	Unit II: Applications of Derivatives (8 hrs) 2.1 Mean value theorems: Rolle's theorem, Lagrange's Theorem (Geometrical interpretation and verification) and applications 2.2 Higher order mean value theorem: Taylor's Series, Maclaurin's Series expansion of function. 2.3 Asymptotes to Cartesian curves up to four degrees. 2.4 Curve tracing in Cartesian form and parametric form 2.5 Curvature
• Evaluate Proper and improper integrals.	Unit III: Integral Calculus (6 hrs) 3.1 Introduction 3.2 Review on Indefinite Integral and fundamental theorem

	of integral calculus. 3.3 Definite integral and its properties 3.4 Improper Integrals; comparison test. 3.5 Reduction formula, Beta Gamma functions
Evaluate arc length, area, and volume through integration.	Unit IV: Application of Integral (6 hrs) 4.1 Application of integrals for finding area beneath a curve and between two curves and arc length 4.2 Surface and volume of solid of revolution in the plane for Cartesian and parametric curves.
Understand and compute partial derivatives with the concept of total differentials.	Unit V: Partial Differentiation (3 hrs) 5.1 Introduction 5.2 Partial Derivatives 5.3 Homogeneous function and Euler's theorem for the function of two and three variables 5.4 Total Derivatives and Differentiation of Implicit functions.
Define extreme value and compute its value for two and three variables through partial derivatives.	Unit VI: Application of Partial Differentiation (4 hrs) 6.1 Extrema of functions of two and three variables. 6.2 Lagrange's method of undetermined Multipliers (up to 2 multipliers)
Able to solve first order differential equations.	Unit VII: First Order Ordinary Differential Equations (6 hrs) 7.1 Review of separable, homogeneous and exact differential equation with engineering applications 7.2 Linear, Bernoulli equation and Riccati's equation with engineering application. 7.3 Mathematical modeling of engineering problems using first order equation.
Solve the second order differential equations in relation to engineering problems.	Unit VIII: Second Order Ordinary Differential Equations (7 hrs) 8.1 Second order Homogeneous ODE with constant and variable coefficients, Euler-Cauchy equation. 8.2 Existence and uniqueness of solutions, Wronskian and general solutions for solving ODE. 8.3 Non-homogeneous second order ODE and Solution by undetermined coefficients and variation of parameters and engineering application

Note: The figures in the parentheses indicate the approximate periods for the respective units.

5. List of Tutorials

Tutorial work covers the work to be done in tutorial. This will enable the students to compute the mathematics problem under the supervision of the course leader. The major tutorial works are as follows:

- Problems on Limit and continuity (1 hr)
- Show that differentiability implies continuity but the converse may not be true (1 hr)
- Evaluation of higher order derivatives by Leibnitz method (1 hr)

4. Problems in Mean value theorems: Rolle's theorem, Lagrange's Theorem (1 hr)
5. Expand the functions through Taylor's Series, and Maclaurin's Series (2 hrs)
6. Evaluation of Asymptotes to Cartesian curves. (2 hrs)
7. Trace Curve for the equations in Cartesian form and parametric form (2 hrs)
8. Problems in Curvature (1 hr)
9. Evaluation of Indefinite Integrals, Definite integrals, Improper Integrals; (2 hrs)
10. Deduce Reduction formula, and solve problems related to Beta Gamma functions (2 hrs)
11. Evaluation of area, arc length, Surface and volume of solid of revolution in the plane for Cartesian and parametric curves. (3 hrs)
12. Prove Euler's theorem for the function of two and three variables (1 hr)
13. Calculate total derivatives and differentiation of Implicit functions. (1 hr)
14. Evaluation of Extrema of functions of two and three variables and Lagrange's method of undetermined Multipliers (up to 2 multipliers) (2 hrs)
15. Solution of separable, homogeneous and exact differential equation Linear, Bernoulli equation and Riccati's equation with engineering applications (2 hrs)
16. Mathematical modeling of engineering problems using first order equation. (1 hr)
17. Solve second order homogeneous ODE with constant and variable coefficients, Euler-Cauchy equation. (3 hrs)
18. Solve non-homogeneous second order ODE by undetermined coefficients and variation of parameters in engineering application (2 hrs)

6. Evaluation System and Students' Responsibilities

Evaluation System

Internal evaluation is done as follows:

External Evaluation	Marks	Internal Evaluation	Weight	Marks
Semester-End Examination	50	Attendance & Class Participation	10%	
		Assignments	20%	
		Presentations/Quizzes	10%	
		Term exam	60%	
		Total Internal		50
Full Marks: 50 + 50 = 100				

Students' Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Books

1. Stewart, J. *Calculus, Early Transcendental*. India; Cengage Learning.
2. Kreyszig, E. *Advance Engineering Mathematics*, New Delhi: John Wiley and Sons Inc.

References

1. Thomas, G. & Finney, R. *Calculus and Analytical Geometry*. New Delhi: Narosa Publishing House.
2. Mishra, P., Mishra, R., Mishra, V. P., & Mishra, M. *Advance Engineering Mathematics*. New Delhi: V. P. Mishra Publication.
3. Dass, H. K. & Verma R. *Higher Engineering Mathematics*. New Delhi: S Chand Publishing.

Pokhara University
Faculty of Science and Technology

Course No.: xxx xxx
Course Title: Applied Physics (3-1-2)
Nature of the Course: Theory and Practical
Year: First
Level: Bachelor

Full Marks: 100
Pass Marks: 45
Time per Period: 1 hour
Total Periods: 45
Program: BE

1. Contents in Detail

Specific Objectives	Contents
Understand mechanical oscillations, solve problems related to different types of oscillation, familiar with the scope in various engineering fields.	Unit I: Mechanical Oscillation (6 hrs) 1.1 Free oscillation, Damped oscillation and Forced oscillation (Physical meaning and equations). 1.2 Compound pendulum, Minimum and maximum time period in compound pendulum, Interchangeability of point of suspension and point of oscillation in compound pendulum, Torsion pendulum, Determination of modulus of rigidity of material using torsion pendulum.
Solve and analyze the problems related to waves	Unit II: Wave Motion (4 hrs) Introduction of wave, wave velocity and particle velocity, types of waves and their applications, Speed of wave in stretched string, energy, power and intensity of plane progressive wave, standing wave and resonance, sonometer.
Solve the problems related to reverberation in different units of building. Solve the problems related to ultra sound.	Unit III: Acoustics (4 hrs.) 3.1 Classification of sound waves, Acoustics of building, Reverberation of sound, absorption coefficient, Noise pollution and its control, Sound insulation, Sabine equation. 3.2 Introduction, production and applications of ultrasonic wave. Ultrasonic method in non-destructive testing.
understand the use of lasers in engineering sciences and solve problems related to laser and fiber optics. Apply the concept of optical fibers in communication system and sensors.	Unit IV: Photonics (6 hrs.) 4.1 Laser: Introduction of laser, Principles of generation of laser light (induced absorption, spontaneous emission, stimulated emission, population inversion, pumping, metastable state), He-Ne laser, Semiconductor laser, Applications of laser. 4.2 Fiber optics: Introduction, Types of optical fiber,

	Principle of propagation of light wave through optical fiber (Acceptance angle), Numerical aperture, Applications of optical fiber in communications, Optical fiber sensors.
Evaluate the capacity of capacitors to store energy with and without dielectrics. Solve problems related to electrostatics.	Unit V: Capacitor and Dielectric (6 hrs.) 5.1 Capacitor: Introduction, Types of capacitors, Charging and discharging of capacitor. 5.2 Dielectric: Introduction, Dielectric constant, electric flux density, Polarization, Polarization in free space, Gauss law in dielectric, Electronic and Ionic polarization (Clausius-Mossotti equation).
Deal with interaction between electric field and magnetic field on matter. Analyze the relationship between electric field, magnetic field and speed of wave.	Unit VI: Electromagnetism (6 hrs.) 6.1 EM Oscillation: LC oscillation, Damped LCR oscillation, Forced electromagnetic oscillation, resonance and quality factor 6.2 EM waves: Maxwell equations in integral form, Conversion of Maxwell's equations in differential form, Continuity equation, Relation between electric field, magnetic field and speed of light, wave equations in free space, verification of light wave as an electromagnetic wave, Wave equation in dielectric medium
Apply principles of quantum mechanics to investigate the observables on known wave functions. Solve the problems related to particle wave using Schrodinger's wave equations.	Unit VII: Quantum Mechanics (5 hrs.) Inadequacy of classical mechanics, Importance of quantum mechanics, Matter wave (de-Broglie equation), Wave function and its significance, Energy and momentum operator, Time independent and time dependent Schrodinger wave equations, Application of Schrodinger wave equation for the electron in metal, Normalized wave function describing the motion of an electron inside in an infinite potential well.
Acquainted with the laws of thermodynamics and applications. Solve the problems related to thermodynamics and heat transfer.	Unit VIII: Fundamentals of Thermodynamics and Heat Transfer (8 hrs.) 8.1 Concepts and definition: applications of thermodynamics, properties and state of substance, thermodynamics properties and types, processes (definition, characteristics and examples): reversible and irreversible process. 8.2 Laws of thermodynamics: first law of thermodynamics, first law for closed system, internal and stored energy, joules law, enthalpy, specific heat, application of first law for closed system, Related problems on closed system, second law of

	thermodynamics, heat engine (four components of refrigerator and heat pump, COP of refrigerator and heat pumps), Kelvin-Planck and Clausius statement of second law. 8.3 Heat transfer: modes of heat transfer (conduction, convection and radiation), statement and assumption of Fourier's law of thermal conductivity, one dimensional steady state heat conduction through plane wall, basic laws of radiation (Emissive power and emissivity, Stefan-Boltzmann's law), Concept of black bodies.
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Student Responsibilities

Each student must secure at least 45% marks in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) and the student will not be eligible to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

2. Prescribed Books and References

Text Books

1. Halliday, D., Resnick, R., & Walker. J. *Fundamental of Physics*. John Wiley and Sons. Inc.
2. Young, H. D. & Freedman. R. A. *Sears and Zemansky's University Physics*. 2009.
3. Howel, J. R. & Buckius, R. O. *Fundamentals of Engineering Thermodynamics*. McGraw-Hill Publishers.

References

1. Reitz, J., Milford, F.J. & Christy, R.W. *Foundations of Electromagnetic Theory*, 1996.
2. David, J. Griffiths, *Introduction to Electrodynamics*, Prentice Hall of India Private Limited, New Delhi, 2008
3. Van Wylen, G. J. and Sonntag, R. E., *Fundamentals of Classical Thermodynamics*, Wiley Eastern Limited, New Delhi, 1989
4. Malik, H. K., Singh, A. K., *Engineering Physics*, Tata McGraw Hill Education Private Ltd., 2010.
5. Arora, C. L. (2020). *B. Sc. Practical Physics*. S. Chand Publishing.
6. Mathur, D. S., *Mechanics*, S. Chand and Company Ltd., 2003.
7. Subrahmanyam, N., Lal, B., *A text book of Optics*, S. Chand and Company Ltd., 2005.
8. Tiwari, K. K. *Electricity and Magnetism*, S. Chand and Company Ltd., 2001
9. Murugesan, R. & Sivaprasath, K., *Modern Physics*, S. Chand and Company Ltd., 2009.